

Prathyusha Dodda

Data Analyst

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Summary

- Experienced **Data Analyst** having around 5 years of experience with a proven track record of delivering data-driven insights that enhance business performance across banking, and IT domains.
- Proficient in **SQL (advanced queries, stored procedures, optimization)**, **Python (pandas, NumPy, matplotlib, Seaborn)**, and **R** for complex data manipulation, analysis, and statistical modeling.
- Skilled in **data visualization** using **Power BI, Tableau, QlikView**, and **Looker**, producing executive-ready dashboards and interactive KPI reports.
- Expertise in **data engineering** and **ETL processes** (Snowflake, Redshift, Azure Synapse, PostgreSQL, Airflow, Data Factory, AWS Glue) with strong big data experience using **Apache Spark/PySpark**.
- Adept at **predictive analytics, time-series forecasting, regression/classification models**, and **A/B testing** to drive marketing, operational, and strategic decisions.
- Strong knowledge of **cloud platforms** (AWS S3, Lambda, RDS, Azure Data Factory, Databricks, Blob Storage, GCP BigQuery) for scalable data solutions.
- Experienced in **data governance, security, and compliance** — implementing GDPR/HIPAA standards, IAM policies, role-based access controls, and data lineage documentation.
- Collaborative communicator and stakeholder partner with expertise in **Agile/Scrum, Jira**, and **Confluence**, ensuring projects are delivered on time, within scope, and to stakeholder expectations.

Technical Skills

Skill Category	Technical Skills & Tools
Data Analysis & Reporting	SQL (complex queries, stored procedures, CTEs, performance tuning), Excel (Pivot Tables, Power Query, VBA), R, Python (pandas, NumPy, SciPy, Jupyter, Matplotlib, Seaborn, Plotly), Statistical modeling & hypothesis testing, Advanced data validation & cleansing
Data Visualization & BI	Tableau, Power BI, Google Data Studio, QlikView, Looker, IBM Cognos Analytics, Sigma Computing, DAX, Custom Calculations
Data Engineering & ETL	Data Warehousing (Snowflake, Redshift, Azure Synapse, BigQuery, PostgreSQL, MySQL, MongoDB), ETL (Apache Airflow, Azure Data Factory, Informatica, AWS Glue), Data Integration (REST API ingestion, Kafka), Data Lake architectures, Schema mapping, Data wrangling, Feature engineering
Big Data Technologies	Apache Spark, PySpark, Hadoop, NoSQL databases (MongoDB, Cassandra), Cloud-native data lakes
Cloud Platforms	AWS (S3, EC2, RDS, Glue, Redshift, Lambda, QuickSight), Azure (Data Factory, Databricks, Blob Storage, Synapse), Google Cloud (BigQuery, Cloud Storage), Containerization (Docker basics), Version control (Git, Bitbucket)
Machine Learning & Automation	Regression/classification/clustering, Time series analysis, Predictive analytics, Scikit-learn, TensorFlow (basics), RapidMiner, Alteryx, A/B Testing, Automated reporting (Power Automate, Python scripting), Automated data cleaning
Statistical & Quantitative	Hypothesis testing, Confidence intervals, ANOVA, Descriptive & inferential statistics, Monte Carlo simulation
Data Collection & Cleansing	Data cleaning, Preprocessing (missing values, duplicates, outliers), Data profiling, Anomaly detection, Data mining
Data Governance & Security	Data privacy (GDPR, HIPAA compliance), Data integrity checks, Audit trails, Access management (AWS IAM, Azure AD, SAS tokens), Secure data transfers, Data quality monitoring, Data lineage management
Business Intelligence	Real-time reporting systems, KPI dashboard development, Market basket analysis, Cohort analysis, Trend analysis, Self-service BI environments, Natural language search analytics (ThoughtSpot)
Collaboration & PM Tools	Agile/Scrum, Jira, MS Project, Documentation (BRDs, SRS, Process Flow diagrams, ER diagrams, BPMN, UML diagrams), Confluence, Change management tools (ServiceNow)
Industry-Specific Platforms	ERP (SAP, Oracle, Microsoft Dynamics), CRM (Salesforce, Zoho), Healthcare Data Systems, Banking Data Platforms

Professional Experience

Data Analyst Simmons Bank, (Remote) USA	Apr 2025 – Current
Engineered SQL-based ETL pipelines in Snowflake, AWS Glue, and Airflow to consolidate multi-source financial transaction, credit, and portfolio data, reducing reporting lag by 40% and enabling near real-time insights for treasury and finance teams. Developed Tableau dashboards for loan performance, delinquency trends, payment defaults, and revenue leakage, improving executive visibility into risk exposures and contributing to a 12% reduction in reconciliation issues. Orchestrated financial data workflows using AWS S3 (data lake), Glue (ETL), and Lambda (serverless automation) to standardize ingestion of transaction and credit bureau feeds, improving refresh frequency and supporting near real-time fraud monitoring. Collaborated with compliance and risk teams to implement standardized regulatory reporting frameworks (SOX, Basel III, Dodd-Frank) documented via Lucidchart and Confluence, improving compliance adherence by 18% and reducing audit escalations. Built Python-driven financial models leveraging Pandas, NumPy, and scikit-learn for credit risk segmentation, probability of default (PD), and loss given default (LGD), improving predictive accuracy of delinquency detection by 22%. Authored BRDs, SRS, and KPI definition guides to ensure consistent financial reporting standards, strengthening data governance, audit readiness, and SOX compliance. Conducted A/B testing and statistical modeling using R and Python to assess engagement with digital banking campaigns, driving a 6% uplift in cross-sell adoption of financial products. Performed advanced SQL profiling and reconciliation scripts across general ledger, loan servicing, and transaction systems, reducing cycle times for month- and quarter-end close by 28%. Integrated external datasets (credit bureau, AML/KYC feeds) into the warehouse using Azure Synapse and AWS Redshift, optimizing schema design and improving scalability of risk and compliance reporting. Partnered with Finance, Treasury, and Risk teams to design dashboards monitoring net interest margin (NIM), liquidity ratios, cost variance, and capital adequacy KPIs, strengthening financial planning and executive reporting. Implemented automated exception handling scripts in Python and AWS Lambda for reconciliation processes, reducing transaction matching errors by 25% and improving monthly and quarterly close accuracy.	

Data Analyst LTI MindTree, India	Sep 2021 – Dec 2023
Designed and implemented SQL-based financial reporting solutions across Oracle and SQL Server, automating reconciliation workflows and reducing month-end discrepancies by 21%, improving reporting accuracy and audit readiness. Built and orchestrated ETL pipelines using Azure Data Factory (ADF) and Informatica PowerCenter to integrate data from multiple financial systems (ERP, CRM, and payment gateways), cutting manual preparation time by 35% and accelerating BI delivery. Developed and published interactive Power BI dashboards connected to Snowflake on Azure, tracking revenue cycle KPIs such as AR aging, cost variance, and SLA adherence, which improved executive decision-making speed and financial visibility. Automated exception handling and data validation scripts using Python (Pandas, NumPy, PyODBC) to detect anomalies in high-volume financial transactions, reducing reconciliation errors by 25%.	

- **Conducted root-cause analysis of financial reporting bottlenecks** by leveraging **advanced SQL queries (window functions, CTEs, dynamic pivoting)** and ETL log tracing, leading to a **28% reduction in reporting delays**.
- **Partnered with finance and compliance teams** to define KPIs, create standardized **BRDs, SRS, and BPMN workflows**, and support monthly operational reporting, reducing manual report preparation by **30%**.
- **Validated financial data migration** from legacy warehouses to **Azure SQL Database and Snowflake on AWS**, writing reconciliation queries to ensure **99.9% data integrity** and compliance with **SOX & GDPR standards**.
- **Implemented financial forecasting and resource optimization models** using **R and Python (scikit-learn, statsmodels)** for regression and time-series analysis, which improved **budgeting accuracy and resource utilization by 12%**.
- **Supported cloud migration initiatives** by optimizing SQL queries, indexing strategies, and **Azure Synapse Analytics** performance tuning, enabling scalable BI workloads across financial reporting systems.
- **Conducted compliance and audit support** for financial processes, ensuring strict adherence to **HIPAA (health-related finance), SOX, and GDPR** regulations through secured access controls and encrypted data flows.
- **Deployed financial data pipelines in Azure Data Lake Storage (ADLS)** to centralize structured and semi-structured financial datasets, reducing siloed reporting and supporting advanced analytics.
- **Integrated Azure Key Vault with ETL workflows** to secure sensitive credentials and financial data transfers, improving audit compliance and data protection.
- **Optimized cost management dashboards** in Power BI by integrating Azure Cost Management APIs, providing real-time tracking of cloud resource spend against financial budgets.
- **Implemented CI/CD pipelines for BI deployments** using **Azure DevOps**, reducing dashboard release cycles by **40%** and ensuring faster delivery of financial reporting enhancements.

Data Analyst Hexaware Technologies, India	Jan 2020 – Aug 2021
• Designed and automated SKU-level data pipelines by extracting and processing sales and inventory data using SQL, Excel, and AWS S3/Athena, enabling daily trend reports that reduced stockouts by 15% and optimized inventory management by 10%. • Developed interactive dashboards in Power BI to visualize pricing sensitivity, seasonal demand shifts, and regional sales trends, allowing marketing teams to adjust campaigns in real time, improving seasonal conversion rates by 11%. • Performed large-scale data quality audits across ERP and cloud systems, detecting duplication and schema drift issues; implemented validation checks that improved master data reliability by 19%. • Analyzed discount impact and footfall trends across store clusters, building AWS-backed comparison reports that improved quarterly demand forecasting accuracy by 20% and supported procurement planning decisions. • Collaborated with business leads on KPI framework development, created reusable templates, and documented data mapping for logistics dashboards, reducing report generation time by 30% and ensuring cross-team alignment. • Authored ER diagrams, process flows, and data lineage documentation in Visio and Confluence, improving onboarding efficiency for junior analysts and cutting integration errors by 25%. • Wrote advanced SQL queries involving joins, subqueries, CTEs, and window functions to generate reconciliation, exception, and sales variance reports across multiple categories. • Optimized stored procedures and queries in Oracle and MySQL, reducing report execution times by 20–25% and supporting faster decision-making during monthly reviews. • Applied Python libraries (Pandas, NumPy, Matplotlib, Seaborn) to clean, manipulate, and visualize transactional datasets; automated preprocessing tasks like handling missing values, outliers, and categorical encoding, improving data readiness for reporting by 30%. • Built Python-based scripts to automate daily ETL tasks such as reading raw CSV/Excel files from AWS S3, applying transformations, and loading curated datasets into staging tables, ensuring data consistency across sources. • Conducted exploratory data analysis (EDA) with Python and SQL, identifying anomalies and sales outliers, which were escalated to business teams, preventing revenue leakage in quarterly reconciliations. • Implemented data cleansing rules for SKU, vendor, and store-level information, ensuring compliance with data governance policies and increasing trust in reporting datasets.	

Education

Anurag Group of Institutions	Jun 2016 – Feb 2021
Bachelor of Technology in Electronics and Communications	

Certifications

- **Certified as Apps.NET Beginner (level 1) Developer**
- **Microsoft Azure Fundamentals (AZ-900)**
- **AWS Certified Cloud Practitioner**
- **AWS Solution Architect - Associate**

Projects

Predictive Analytics for Customer Churn Reduction
• Built a machine learning model in Python (pandas, NumPy, scikit-learn) to predict churn using behavioral and transactional datasets, achieving 85% predictive accuracy after feature engineering, cross-validation, and A/B testing. • Delivered churn risk insights through Power BI dashboards that segmented high-risk customers, enabling targeted retention strategies and projected cost savings in customer acquisition.

Sales & Inventory Forecasting Using Time Series Analysis

- Designed time-series forecasting models (Prophet, statsmodels) on historical sales data to optimize inventory levels, reducing stockouts by 15% and minimizing over-stock expenses.
- Automated data ETL with SQL and Apache Airflow, visualizing forecasts in Tableau for supply chain decision support.

Operational Reporting and Revenue Monitoring

- Created SQL-based reports and Power BI dashboards to track claim denials and payment delays.
- Accelerated issue resolution by 28% and minimized missed revenue by 17% through clear metric tracking.
- Ensured 99.7% data accuracy during system migration by validating record-level outputs.